Machine Learning 2 (APM_50441_EP/DSAIB - 2026-2027). PC5

Exercise 1 (SVD and the reconstruction problem)

Let X be an $n \times p$ matrix. Show that a solution to the following optimization problem

$$\min_{u \in \mathbb{R}^n, v \in \mathbb{R}^p} J(u,v) \qquad \text{with} \quad J(u,v) = \|X - uv^\top\|_F^2$$

with $\|v^\star\| = 1$, is given by

$$v^\star \qquad \text{and} \qquad u^\star = X\, v^\star,$$

where $v^\star$ is the normalized eigenvector associated with λ the largest eigenvalue of $X^\top X$. Furthermore, $\|u^\star\| = \sqrt{\lambda}$.

Exercise 2 (Iterated SVD)

The weighted SVD of order q problem is

$$\min_{U \in \mathbb{R}^{n \times q}, V \in \mathbb{R}^{p \times q}} \|X - UV^\top\|_W^2$$

where

$$\|M\|_W^2 = \sum_{i,j} W_{i,j} M_{i,j}^2.$$

We assume here that W is a mask, that is $W_{i,j} \in \{0,1\}$.

Let $\odot$ be the component-wise multiplication, the iterated SVD algorithm is given by

- Start by an initial factorization $U_0 V_0^\top$.
- Iterate T times:
  - Compute the completed matrix $R_t = W \odot X + (1 - W) \odot (U_t V_t^\top)$ where 1 is a matrix filled with 1 everywhere.
  - Use the SVD to obtain the best rank q approximation (truncated SVD) of R_t, written $U_{t+1} V_{t+1}^\top$.
- Use the last factorization $U_T V_T^\top$.

We let $\mathcal{L}(U,V) = \|X - UV^\top\|_W^2$ and define

$$\mathcal{L}^+(U,V,U',V') = \|W \odot X + (1 - W) \odot U'V'^\top - UV^\top\|^2$$

where $\|\cdot\|$ is the (unweighted) Frobenius norm.

1. Verify that $\mathcal{L}(U,V) \leq \mathcal{L}^+(U,V,U',V')$ and $\mathcal{L}(U,V) = \mathcal{L}^+(U,V,U,V)$
2. Prove that $\mathcal{L}(U_{t+1}, V_{t+1}) \leq \mathcal{L}(U_t, V_t)$. You may use the fact that the SVD can be used to minimize the problem

$$\min_{U \in \mathbb{R}^{n \times q}, V \in \mathbb{R}^{p \times q}} \|X - UV^\top\|^2.$$

3. *Soft-Impute.* Instead of a fixed rank q, Soft-Impute penalizes the nuclear norm $\|Z\|_* = \sum_k \sigma_k(Z)$ (the sum of the singular values, a convex surrogate of the rank) and solves $\min_Z \frac{1}{2}\|X - Z\|_W^2 + \lambda \|Z\|_*$. Admit that the solution of $\min_Z \frac{1}{2}\|R - Z\|^2 + \lambda \|Z\|_*$ is the *soft-thresholded SVD* $S_\lambda(R) = \sum_k (\sigma_k - \lambda)_+ u_k v_k^\top$ of $R = \sum_k \sigma_k u_k v_k^\top$. Show that the iteration $Z_{t+1} = S_\lambda\left(W \odot X + (1 - W) \odot Z_t\right)$ decreases the objective, by the argument of the previous question. Show finally that $\|Z\|_* = \min_{UV^\top = Z} \frac{1}{2}\left(\|U\|^2 + \|V\|^2\right)$ and relate Soft-Impute to the penalized ALS of the next exercise.

Exercise 3 (**Alternating Least Squares**)

The penalized weighted SVD of order q problem is

$$\min_{U\in\mathbb{R}^{n\times q},V\in\mathbb{R}^{p\times q}} \|X-UV^\top\|_W^2+\lambda\|U\|^2+\lambda\|V\|^2$$

with $\lambda\geq 0$, $\|\cdot\|$ the Frobenius norm and

$$\|M\|_W^2=\sum_{i,j} W_{i,j}M_{i,j}^2$$

with $W_{i,j}\geq 0$ (with a $0/1$ mask, $W_{i,j}$ and $W_{i,j}^2$ coincide; the slides write the weights $W_{i,j}^2$).

To solve it using the penalized Alternating Least Squares (ALS) the following sequence has to be implemented: initialize U and V with the SVD of the mean-filled matrix (the missing entries replaced by the mean rating)

loop
compute U minimizing $\|X-UV^\top\|_W^2+\lambda\|U\|^2$ with V fixed
compute V minimizing $\|X-UV^\top\|_W^2+\lambda\|V\|^2$ with U fixed

1. Show that
$$\min_U \|X-UV^\top\|_W^2+\lambda\|U\|^2 \text{ with } V \text{ fixed}$$
boils down to solving n least squares problems of the form
$$\min_{u_i\in\mathbb{R}^q} \|x_i-Vu_i\|_{D_i}^2+\lambda\|u_i\|^2, \qquad \text{for } i=1,\dots,n$$
with $x_i=X_{i,\cdot}^\top$ a vector of $\mathbb{R}^p$, $D_i=\mathrm{diag}(W_{i,\cdot})$ the diagonal matrix with $W_{i,j}, j=1,\dots,p$ on the diagonal, $\|x\|_D^2=\sum_j D_{j,j}x_j^2$, and whose solution is
$$u_i=(V^\top D_iV+\lambda I)^{-1}V^\top D_ix_i,$$
where I is the identity matrix of dimension q.

2. Show that
$$\min_V \|X-UV^\top\|_W^2+\lambda\|V\|^2 \text{ with } U \text{ fixed}$$
boils down to solving p least squares problems of the form
$$\min_{v_j\in\mathbb{R}^q} \|x_j-Uv_j\|_{D_j}^2+\lambda\|v_j\|^2, \qquad \text{for } j=1,\dots,p$$
with $x_j=X_{\cdot,j}$ a vector of $\mathbb{R}^n$ and $D_j=\mathrm{diag}(W_{\cdot,j})$ the diagonal matrix with $W_{i,j}, i=1,\dots,n$ on the diagonal, and whose solution is
$$v_j=(U^\top D_jU+\lambda I)^{-1}U^\top D_jx_j,$$
where I is the identity matrix of dimension q.

3. *Biases.* The biased factorization of the lecture predicts $\hat{X}_{ij}=\mu+b_i+c_j+u_i^\top v_j$ (global mean, user bias, item bias). Show that the update of (b_i,u_i) with V and c fixed is again a ridge regression, by augmenting the vectors: $\tilde{u}_i=(b_i,u_i^\top)^\top$ and $\tilde{v}_j=(1,v_j^\top)^\top$. For $U=V=0$, compute the regularized user bias b_i (with $c=0$) and interpret it.

Exercise 4 (**Ranking metrics**)

A recommender proposes a ranked list of K items to a user. With $\mathrm{rel}_k\in\{0,1\}$ the relevance of the item at rank k and R the number of relevant items of the user, P@$K=\frac{1}{K}\sum_{k\leq K}\mathrm{rel}_k$, R@$K=\frac{1}{R}\sum_{k\leq K}\mathrm{rel}_k$, DCG@$K=\sum_{k\leqslant K}\frac{\mathrm{rel}_k}{\log_2(k+1)}$, NDCG@$K$ = DCG@K/IDCG@K (IDCG: the DCG of the ideal ranking) and the reciprocal rank is $1/$(rank of the first relevant item).

1. A user has $R=3$ relevant items, and a recommender ranks them at positions 1, 4 and 6 of its list. Compute P@5, R@5, NDCG@5 and the reciprocal rank.

2. Construct a small example (one user, 4 items, ratings from 1 to 5) where a model A has a smaller RMSE than a model B on the observed ratings, but a worse ranking (NDCG) of the items. What does it show about the evaluation of recommender systems?

Exercise 5 (Implicit feedback: weighted ALS and BPR)
With implicit data (clicks, views), only positive interactions are observed. Hu, Koren and Volinsky (2008) minimize $\sum_{i,j} c_{ij}(p_{ij} - u_i^\top v_j)^2 + \lambda(\|U\|^2 + \|V\|^2)$ over *all* the pairs, with the preference $p_{ij} = \mathbf{1}_{r_{ij}>0}$ and the confidence $c_{ij} = 1 + \alpha r_{ij}$ (where r_{ij} counts the interactions).

1. Using the previous exercise, write the ALS update of u_i, and show that $V^\top C_i V = V^\top V + V^\top(C_i - I)V$ (with $C_i = \mathrm{diag}(c_{i\cdot})$) allows computing it in $O(n_i q^2 + q^3)$ operations, where n_i is the number of items of the user, once $V^\top V$ is computed.

2. Bayesian Personalized Ranking (BPR) maximizes $\sum_{(i,j,k)} \log \sigma(\hat{x}_{ij} - \hat{x}_{ik})$ over triples where user i interacted with j but not with k (sampled: *negative sampling*), with $\hat{x}_{ij} = u_i^\top v_j$ and σ the logistic function. Compute the gradient with respect to u_i, and explain why this criterion is a smooth surrogate of the AUC of the ranking of each user.

Exercise 6 (Exposure bias and inverse propensity scoring)
The logged data of a recommender only contain the feedback of the items that were *shown*. Let $O_{ij} \in \{0, 1\}$ indicate that item j was shown to user i, with probability (propensity) $\pi_{ij} > 0$, independently of the feedback Y_{ij}. We want to estimate the average loss $L = \frac{1}{np}\sum_{i,j} \ell(Y_{ij}, \hat{Y}_{ij})$ over all the pairs.

1. Show that the naive average over the shown pairs is biased in general, and that the IPS estimator $\hat{L} = \frac{1}{np}\sum_{i,j} \frac{O_{ij}}{\pi_{ij}} \ell(Y_{ij}, \hat{Y}_{ij})$ is unbiased.

2. Why does the variance of $\hat{L}$ explode when some propensities are small? Propose a remedy.

Exercise 7 (Text representation) 1. *tf-idf.* For three documents d_1 = "the cat eats the fish", d_2 = "the dog eats", d_3 = "the fish swims", compute the tf-idf weights $\mathrm{tf}(w, d)\log(N/\mathrm{df}(w))$ (with $N = 3$ and the raw counts as tf) of the words of d_1. Which words are the most characteristic of d_1?

2. *Word2vec as a factorization.* The skip-gram with negative sampling maximizes, for each pair (word w, context c), $\#(w, c)\log\sigma(u_w^\top v_c) + k\,\#(w)\frac{\#(c)}{T}\log\sigma(-u_w^\top v_c)$, where k is the number of negative samples, drawn from the empirical distribution of the contexts, and T the total number of pairs. Considering $x = u_w^\top v_c$ as a free variable, show that the optimum is $x = \mathrm{PMI}(w, c) - \log k$, where $\mathrm{PMI}(w, c) = \log\frac{\#(w,c)\,T}{\#(w)\#(c)}$ (Levy and Goldberg, 2014). Relate this to the PPMI + SVD word vectors of the lecture.

3. *Attention.* For a sequence of n tokens $X \in \mathbb{R}^{n\times d}$, self-attention computes $\mathrm{softmax}(QK^\top/\sqrt{d_k})V$ with $Q = XW_Q$, $K = XW_K$, $V = XW_V$ (softmax along the rows). Show that it is permutation-equivariant: permuting the tokens permutes the outputs in the same way. Why are positional encodings needed? If the entries of Q and K are independent, centered, of variance 1, what is the variance of an entry of $QK^\top$, and why is it divided by $\sqrt{d_k}$?

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